

Introduction to Clustering

(IART Lab Exercises)

April 23, 2026

Exercise 1

Use the k-means clustering method to analyze the behavior of customers of a catalog sales company.

a) Load the data

Load the sheet “catalog” that is available in the excel file provided.

b) Generate a clustering

Use the k-means algorithm. Configure the algorithm to generate two clusters.

c) Interpret the clustering

1. What is the meaning of the values associated with each cluster?
2. Is it possible to assign a short description of the profile of the typical customer in each cluster?

Exercise 2

Explore the results obtained in the previous exercise in more detail.

a) Variable selection

1. Discuss whether the variable “ID” should be considered.
2. Generate clusterings with and without this variable. What is the difference?
3. Discuss whether there are additional variables that should be considered.

b) Number of clusters

Vary the number of clusters that are generated and interpret the results. What changes in the clusters and corresponding profiles?

c) Data preparation

Consider the clusterings obtained so far.

1. Are there variables that seem more important than the others? Which ones? Why do you think this happens?
2. Normalize the data: what does it do?
3. Run the clustering: are the profiles still understandable?
4. Denormalize the centroids. What is the impact of normalization on the generated clusters?

Exercise 3

Repeat the exercises for the data in the “wholesales customers” sheet of the same excel file.

Exercise 4

Repeat the exercises for the data in the “credit” sheet of the same excel file.